Skyler Lang

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EDUCATION

University of Illinois Urbana-Champaign, B.S. Computer Engineering — Expected May 2027

EXPERIENCE

Network Automation Intern, Lenovo *Morrisville, NC* — Jun – Jul 2026

- Built a Python tool that converts AI-cluster network designs into deployable containerlab/SONiC digital twins, taking a design to a fully routed fabric in ~5 min with one command
- Cut fabric configuration time 75% (8 → 2 min) with a bounded thread pool, and added a byte-identical dry-run gate that caught a real regression on its first run
- Designed numbered eBGP across 5 networks with a switch-identity ASN scheme and a position-encoded /31 IP allocator as the single source of truth
- Built 4-layer verification (LLDP, BGP state, routes, TTL-checked ping) that exposed a dead core tier a healthy-looking deploy had missed
- Root-caused 7 upstream design-tool bugs with file/line citations, fixed 4+ in inherited code, and completed the tool's unfinished actualization module

AI Engineer Intern, Jess-Link Products *Taiwan* — Jun – Aug 2025

- Proposed and built the company's first AI optical-inspection system, a YOLO solder-defect detector (99% mAP@0.5, Roboflow-labeled) that flags insufficient-solder joints before UV cure, while rework is still possible
- Ran it on a GPU workstation at the assembly line, where it stayed in production after the internship and was extended to PCB defect inspection
- Built a RAG tool (LangChain, ChromaDB, Ollama) over spec sheets and datasheets so engineers could query part specs in plain English

PROJECTS

ArcVision AI — computer-vision basketball shot tracker | arcvisionai.com — Apr – Sep 2026

- Fine-tuned a 3-class YOLOv8n detector on self-collected video to 0.982 mAP50 and built the Python/OpenCV pipeline with a custom centroid tracker and make/miss classifier
- Traced a 73% accuracy ceiling to classifier logic and ground-truth errors rather than the detector, then fixed both to reach 91% and ~99% on a 40-shot multi-angle set

lifeOS — personal automation platform — 2026

- Built a mobile PWA on Supabase/Postgres (edge functions, pg_cron, RLS) with an LLM planning layer and ~40 SQL functions, including an in-database job scraper and resume-tailoring pipeline

RISC-V Operating System (ECE 391) — 2026

- Wrote VirtIO block device drivers, the virtual-memory subsystem, and filesystem concurrency fixes in C for a RISC-V kernel

Robotic Arm Control (ECE 470) — Spring 2024

- Wrote the Python control software for a UR3 robotic arm, implementing forward and inverse kinematics, path planning, and camera-based block pick-and-place

Micromouse (iRobotics) — 2024 – 2025

- Designed most of the hardware, soldered the board, and wrote the Raspberry Pi firmware for a maze-solving micromouse robot, including motor control

Wireless Guitar System — 2024 – 2025

- Designed the analog front end and digitization chain (NE5534 preamp, MCP3008 ADC over SPI to an ATmega328), validated it on an oscilloscope, and laid out the schematic and PCB in KiCad

SKILLS

Languages: Python, C, C++, Java, SQL, SystemVerilog, JavaScript, Bash

Systems & Embedded: Linux, Docker, containerlab, BGP, SONiC, RISC-V, device drivers, UART, SPI, I2C, Arduino, ESP32, Raspberry Pi

Hardware & ML: KiCad, Cadence Allegro, oscilloscope, Vivado, PyTorch, YOLO, OpenCV, Roboflow

Backend & Tools: PostgreSQL, Supabase, Django, NetBox, LangChain, ChromaDB, Git, GDB